

Project Proposal

● Graded

Group

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Total Points

4.75 / 5 pts

– 0 pts Excellent proposal.

✓ – 0.25 pts Strong proposal, minor areas for improvement

– 0.5 pts Good proposal, some areas for improvement

– 0.75 pts Good proposal, multiple areas for improvement

– 1 pt Click here to replace this description.

- 1 Or vice-versa! Remember, there are other reasons to generalize hierarchically (i.e., the rest of the grammar is hierarchical), so simplicity bias may lead to a preference for the hierarchical generalization
- 2 These seem like reasonable values!
- 3 The training corpus should be not just SVA sentences. What is the source of the Linzen sentences, wikipedia? If so, you can access the same wiki dump to get up to 100M words. Or you may find it's easier ultimately to use the parsed BabyLM data and write your own syntactic filter
- 4 These are relevant papers, but I expect some more discussion about this. And is there any other relevant work looking at noise in the input? Misra and Mahowald, for example, but do look for others
Why compare GPT2 and LSTM specifically? Syntactically biased models might make for a more motivated comparison:
<https://direct.mit.edu/tac/article/doi/10.1162/tac.1.00526/114315>
- 5 <https://aclanthology.org/2020.emnlp-main.375.pdf>
<https://direct.mit.edu/tac/article/doi/10.1162/tac.1.00304/43542/Does-Syntax-Need-to-Grow-on-Trees-Sources-of>
- 6 I would suggest a greater number of runs, if compute permits (e.g., repeat each condition 3x)
- 7 Also, is there a bias towards linear or hierarchical generalization in the completely 50/50 case?
- 8 As in the delta between the minimal pairs, or delta for the grammatical sentence between baseline and noisy conditions? (both are reasonable)
- 9 Ideally, you should manually validate a subset of your corpus to determine whether there are other agreement attraction errors in the corpus (i.e., natural ones that you didn't insert) and whether the inserted/modified sentences are correctly modified
- 10 Why "yet"? The fact that children learn despite noise is key to the argument in favor of innate knowledge
- 11 Strong proposal! See comments below. Looking forward to seeing your findings!

DSC 291 Project Proposal: Robustness to Noise in Grammar Rule Learning

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Course: DSC 291 — Language Models as Cognitive Models, UCSD

Advisor: Alex Warstadt

Date: May 7, 2026



1. RESEARCH QUESTION & MOTIVATION

Children acquire robust grammatical rules despite noisy input, speech errors, non-native speech, and agreement violations in the environment. How resilient are language models to the same kind of noise?

We focus on subject-verb agreement (SVA), the most extensively studied grammatical construction in the LM probing literature. Our central questions are:

1. Can injecting SVA agreement errors into training data delay or prevent a model from acquiring the correct hierarchical agreement rule?
2. Is the counterfactual linear agreement rule equally learnable under noise?
3. Do GPT-2 (transformer) and LSTM (recurrent) architectures differ in their robustness to noise?



This connects directly to a foundational debate in language acquisition. Chomsky's poverty of the stimulus argument holds that linguistic input is too noisy and incomplete for grammar to be learned purely from data, implying some grammatical knowledge must be innate. Yet children demonstrably acquire robust rules despite hearing speech errors, non-native input, and agreement violations. If LMs similarly succeed under noisy training conditions, it suggests grammar may be more learnable from input than traditionally assumed. If they fail, it raises the question of what children have that LMs lack, whether that is innate structure, embodied experience, or something else entirely.



2. RELATED WORK

- Linzen et al. (2016). Assessing the ability of LSTMs to learn syntax-sensitive dependencies. TACL. [Foundational SVA evaluation; source of the rnn_agreement dataset]
- Gulordava et al. (2018). Colorless green recurrent networks dream hierarchically.

- NAACL. [Hierarchical vs. linear rule generalization in LSTMs]
- BLiMP (Warstadt et al., 2020). A challenge set for evaluating grammatical knowledge.
- TACL. [Minimal pair evaluation paradigm]
- SyntaxGym (Gauthier et al., 2020). [Targeted syntactic evaluation via surprisal]

3. EXPERIMENTAL DESIGN

--- Data ---

Base corpus: Linzen rnn_agreement dataset
https://github.com/TalLinzen/rnn_agreement

We will construct 5 training corpora by injecting SVA agreement errors at increasing rates:

Condition	Error rate
Baseline	0%
Low	0.4%
Medium-low	2%
Medium-high	10%
High	50%

Error injection: for each sampled sentence, flip subject-verb agreement with probability p (the noise rate). This produces corrupted corpora that otherwise preserve all other syntactic and lexical properties.

--- Architectures ---

Both architectures will be trained from scratch (not fine-tuned) on each noise condition:

1. GPT-2-small (primary)
 - a. Transformer-based autoregressive LM
 - b. Professor's recommended architecture
 - c. Evaluated on next-token surprisal and minimal pair accuracy
2. LSTM (secondary)
 - a. Recurrent LM matching Linzen et al. (2016) setup
 - b. Grounds our results in prior literature for direct comparison
 - c. Trains significantly faster; can run in parallel with GPT-2

Total model runs: 5 noise levels $\times$ 2 architectures = 10 training runs

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--- Evaluation Metrics ---

1. Primary: Minimal pair accuracy on SVA test sets
(does the model assign higher probability to the grammatical sentence?)
2. Secondary: Surprisal scores for subject-verb agreement
(how surprised is the model by the correct verb form?)

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We will also probe whether models acquire the hierarchical rule (sensitive to the true subject, regardless of intervening nouns) vs. the linear rule (sensitive to the most recent noun before the verb).

--- Compute ---

Primary: UCSD teaching cluster

Backup: Alex Warstadt's lab cluster

4. TIMELINE & MILESTONES

Week	Milestone
6	Proposal due; set up compute environment and repo
7	Data pipeline: construct and verify 5 noise-level corpora; confirm training scripts run end-to-end on baseline
8	Train all 10 models (5 noise $\times$ 2 architectures); collect raw results
9	Evaluation runs; analysis (accuracy curves, rule probing); begin paper draft
10	Project presentations
11	Final paper due 6/11

5. DIVISION OF LABOR

1. Data & Training Infrastructure

- a. Owns area. Sets up compute environment (teaching cluster access, job scripts); coordinates and monitors all training runs [Jackson]
 - b. Builds the noise injection pipeline: takes rnn_agreement corpus, injects SVA errors at each of the 5 rates, validates output distributions [Kelly]
 - c. Implements training scripts for GPT-2-small and LSTM; manages model checkpointing and logging [Sanjana]
2. Evaluation & Analysis
 - a. Owns area. Implements minimal pair accuracy evaluation across all 10 models; aggregates and organizes results [Sanjana]
 - b. Implements surprisal scoring and hierarchical vs. linear rule probing [Kelly]
 - c. Result visualization: accuracy curves by noise level, cross-architecture comparison plots, statistical analysis [Jackson]
3. Writing & Presentation
 - a. Owns area. Coordinates paper structure and section assignments; writes introduction and conclusion [Kelly]
 - b. Writes methods and experimental setup sections (data, noise manipulation, architectures) [Jackson]
 - c. Writes results and discussion sections; creates presentation slides [Sanjana]

6. EXPECTED CONTRIBUTIONS

1. Empirical characterization of how noise rate affects SVA rule acquisition in GPT-2 and LSTMs — including whether there is a critical noise threshold beyond which acquisition breaks down.
2. Comparison of hierarchical vs. linear generalization under varying noise conditions — does noise push models toward simpler (linear) strategies? 
3. Cross-architecture comparison (transformer vs. RNN) under identical noise conditions — do transformers and LSTMs differ in their robustness to noisy grammar input?

7. DISCUSSION OF POSSIBLE OUTCOMES

We anticipate three possible outcome patterns. First, models may remain robust at low noise rates (0.4–2%) but degrade sharply at higher rates (10–50%), suggesting a critical threshold for input quality beyond which grammar learning breaks down. Second, noise may  disproportionately impair hierarchical rule learning while leaving linear rule performance intact, indicating that noise pushes models toward simpler generalization strategies. Third, transformers and LSTMs may differ in robustness if transformers outperform LSTMs under

No questions assigned to the following page.

noise, it implicates architectural differences; if both fail, it strengthens nativist arguments that input alone is insufficient for grammar acquisition.

